

DŽENAN KREHO

 dzenankreho.github.io

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TECHNICAL SKILLS

Programming languages:

C/C++

MATLAB

Python

Java

Assembly

LabVIEW

PLC programming:

LD

FBD

SFC

ST

Frameworks and libraries:

FreeRTOS

JUnit

CMSIS-DSP

Qt

OpenCV

Tools:

Simulink

Git

CMake

Linux

Doxygen

IDEs:

CLion

Visual Studio

PyCharm

IntelliJ IDEA

Code Composer Studio

Keil uVision

MPLAB X

LANGUAGES

Bosnian: **Native**

English: **Fluent**

Swedish: **Elementary**

German: **Elementary**

EDUCATION

Master of Electrical Engineering
with Specialization in Automatic Control and Electronics

University of Sarajevo, Bosnia and Herzegovina

 Oct 2022 – Sept 2024

- Received the University of Sarajevo Silver Badge in acknowledgment of academic achievement.

Master of Science in Computer Science
with Specialization in Embedded Systems

Mälardalen University, Sweden

 Aug 2023 – Jun 2024

- Obtained through the Erasmus+ International Credit Mobility programme.

Bachelor of Electrical Engineering
with Specialization in Automatic Control and Electronics

University of Sarajevo, Bosnia and Herzegovina

 Oct 2019 – Sept 2022

- Received the University of Sarajevo Silver Badge in acknowledgment of academic achievement.

EXPERIENCE

HVDC Control and Protection Engineer

Hitachi Energy

 Jan 2025 – Present

- Analyzing and managing technical customer requirements, and contributing to customer meetings.
- Designing a supervision system for secondary equipment of the converter station.
- Designing a state machine and sequences for converter station startup and shutdown operation.
- Designing a transient fault recorder using hardware and software from iba AG.
- Preparing technical documentation (function descriptions, signal lists, overview diagrams, etc.) and FAT test cases for the above HVDC converter station sub-systems.

Master's Thesis Student

Volvo Construction Equipment

 Jan 2024 – May 2024

- Worked on a Master thesis focused on functionality allocation, timing analysis, and schedulability analysis of multi-core ECUs.
- Developed and tested a tool in the C programming language that implements the proposed analyses and techniques.

Student Researcher

University of Sarajevo, Faculty of Electrical Engineering

 Nov 2021 – Feb 2023

- As part of a research project, worked on the GUI component of a real-time simulator for smart grid networks.
- Wrote C++ code that uses the Windows App SDK, GTK 4, and 3D rendering APIs: OpenGL and Vulkan.
- Performed performance testing, documented the results, and documented the development process in the form of beginner tutorials to assist other project team members in the future.

Undergraduate Teaching Assistant

University of Sarajevo, Faculty of Electrical Engineering

 Mar 2022 – Jun 2022

- Conducted, assisted students with, and graded laboratory exercises for the courses: C++ Programming and Lab Work in Data Acquisition and Control.